

INTRODUCTION TO INFO. & COMM. TECHNOLOGY

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Hexadecimal to Binary

- Convert each Hexadecimal digit into 4-digit binary code

- Example:

E	1	9	C ₁₆
1110	0001	1001	1100

Binary to Hexadecimal

- Make sets of four digits starting from Right to Left and convert each set into equivalent hexadecimal digit

- Example

1110101000001111₂

1110 1010 0000 1111

E A 0 F → EA0F₁₆

Hexadecimal to Octal

2 A 1 4 9₁₆

0010 1010 0001 0100 1001 4-bit code for each

Make sets of 3-bits from R to L and convert into decimal

0 0 1 0 1 0 1 0 0 0 1 0 1 0 0 1 0 0 1

00 101 010 000 101 001 001

(0 5 2 0 5 1 1)₈

Octal to Hexadecimal

612753 in octal system = ? In hexadecimal system

Convert each digit in 3-bit binary code

6	1	2	7	5	3
110	001	010	111	101	011

Make sets of 4-bits from R to L

1 1 0 0 0 1 0 1 0 1 1 1 1 0 1 0 1 1

Convert each set into decimal system

00 1 1	0 0 0 1	0 1 0 1	1 1 1 0	1 0 1 1
3	1	5	14	11

3 1 5 E B

Addition in Binary No. Systems

- $0 + 0 = 0$
- $0 + 1 = 1$
- $1 + 0 = 1$
- $1 + 1 = 10$

Binary Arithmetic

- Binary Addition

1. $00111101 + 00111010$
2. $00101101 + 00110010$
3. $01100010 + 01010111$
4. $01010111 + 00111101$
5. $00101001 + 11100101$

Find Answers and verify from your calculator

Decimal Subtraction ?

- $7012 - 678$?

Decimal Subtraction ?

- $7012 - 678$?

- 6334

Binary Subtraction

1. $110 - 10$
2. $101 - 11$
3. $1001 - 11$
4. $1101 - 11$
5. $10001 - 100$

Program -1

- Nippon Paints, a company that produces paint pays an annual bonus to its employees.
- The bonus is calculated by multiplying the employee's annual salary by a bonus rate.
- Develop a program logic that accepts the name of the employee (Name), Annual salary (A_Salary), and the bonus rate(bonus_rate).
- The program displays a greeting message and the amount of bonus earned by the employee

Program - 2

- Design a program logic that will determine the area and perimeter of a rectangle
- Accept the length and width from the user
- Calculate and display the answers using the following formulae:

$\text{Area} = \text{Length} * \text{Width}$

$\text{Perimeter} = 2(\text{Length} + \text{Width})$

Program - 3

- Write a code logic that accepts years of experience for an employee and the salary.
- If the experience is greater than 5 years, bonus is 10% of the salary.
- If the experience is less than 5 years, then bonus is 5% of the salary.

Binary Multiplication

The Binary multiplication obeys the four basic rules:

$$0 \times 0 = 0$$

$$1 \times 0 = 0$$

$$0 \times 1 = 0$$

$$1 \times 1 = 1$$

Binary Multiplication

$$\begin{array}{r} 1000 \\ x 1001 \\ \hline 1000 \\ 0000 \\ 0000 \\ 1000 \\ \hline 1001000 \end{array}$$

Binary Multiplication

```

      01010010 (multiplicand)
    x 01101101 (multiplier)
    -----
          01010010
         00000000
        01010010
       01010010
      00000000
     01010010
    01010010
   00000000
  -----
 010001011101010

```


Binary Multiplication

1. 10×10
2. 100×11
3. 101×10
4. 1011×11
5. 11011×101

Answers

1. 100
2. 1100
3. 1010
4. 10 0001
5. 1000 0111

Binary Division

$$\begin{array}{r} 1000 \quad | \quad \begin{array}{r} 1001 \\ \hline 1001010 \\ -1000 \\ \hline 10 \\ 101 \\ 1010 \\ -1000 \\ \hline 10 \end{array} \end{array}$$

Quotient
Dividend

Remainder

Verifying division

Dividend = Quotient x Divisor + Remainder

$$\begin{aligned} 1001010 &= (1001 \times 1000) + 10 \\ &= 1001000 + 10 \\ &= 1001010 \end{aligned}$$

Division

$$\begin{array}{r}
 1000 \overline{) 1001010} \\
 \underline{-1000} \\
 10 \\
 \underline{101} \\
 1010 \\
 \underline{-1000} \\
 10
 \end{array}$$

$$\begin{array}{r}
 0010 \overline{) 00000111} \\
 \underline{-0010} \\
 00011 \\
 \underline{-0010} \\
 0001
 \end{array}$$

Binary Division

1. $100 / 10$
2. $111 / 11$
3. $1010 / 100$
4. $1101 / 11$
5. $10111 / 10$

Answers

1. 10
2. 10 (R = 0001)
3. 10 (R = 0010)
4. 100 (R = 0001)
5. 1011 (R = 0001)

Ref Chapters

- Book 2
- Chp 5